TDW6323 DATA WRANGLING AND VISUALISATION

Lab Assignment (20%)

A. Description:

In this assignment, you will work in a group (minimum **THREE**, maximum **FOUR** students per group) to demonstrate your data wrangling and visualisation skills using Python. This assignment consists of three parts.

B. Parts:

This assignment consists of three parts: 1. Data Cleaning and Transformation, 2. Feature Selection, and 3. Data Visualisation. Each of these parts is explained as follows:

Dataset used:

- Part 1, Part 2 and Part 3:
  Dataset1- **Diabetic_Data.csv**

Part 1- Data Cleaning and Transformation (10 Marks)

You will be working with a dataset containing various data quality issues, such as missing data, outliers, unstructured data, and more. Your task is to clean, preprocess, and transform the data to make it suitable for visualisation or further analysis and modeling. Please perform the following tasks using Dataset1:

1. Missing Data Handling (4 Marks)

- Display and describe the dataset.
- Inspect the dataset, check the number of rows and cols.
- Transform 'readmitted' column and refine the nan values.
- Identify the columns with missing values, detect missing values and decide on an appropriate strategy to handle them.
- Implement the chosen method to fill the missing data.
- Drop the non-impactful columns.
- Split the numerical and non-numerical attributes for better preprocessing.
- Show that each column with missing values are resolved.

2. Data Cleaning and Transformation (3 Marks)

- Transform target variable to numerical and refine the NaNs, identify and plot the number of readmitted and not readmitted instances.
- Identify the impact of each 'gender', 'age' and 'race' on readmitted class, plot it separately using histogram or a bar graph. Explain the frequency distribution briefly.

- Calculate the average for insulin rate for readmitted instances and group them by categories (‘Down’, ‘No’, ‘Steady’, ‘Up’). Visualise and plot the impact of insulin intake on target class of readmitted by their age group, gender and race. You can use colors, bar graphs or histogram to present the outcome. Explain the graphs briefly.
- Clean up the diagnosis columns, ‘diag_1’, ‘diag_2’ and ‘diag_3’, use fillna keyword and transform the values. Create new categories following the range given below:

```
value>=390 and value<=459 or value==785:
    category = 'Circulatory'
value>=460 and value<=519 or value==786:
    category = 'Respiratory'
value>=520 and value<=579 or value==787:
    category = 'Digestive'
value==250:
    category = 'Diabetes'
value>=800 and value<=999:
    category = 'Injury'
value>=710 and value<=739:
    category = 'Musculoskeletal'
value>=580 and value<=629 or value==788:
    category = 'Genitourinary'
value>=140 and value<=239 :
    category = 'Neoplasms'
value==-1:
    category = 'NAN'
if none above matches:
    category = 'Other'
```

  Plot the frequency distribution using bar graph and explain which category is most readmitted.

- Select 21 drug features listed in the dataset and replace the values ‘No’, ‘Steady’, ‘Up’, ‘Down’ to [0,1,1,1]. Ensure to set the column datatype as integer.
- Create a new col ‘A1Cresult’ and ‘max_glu_serum’ and transform the attributes into integer from string:
    1) 'A1Cresult' = replace ('>7','>8','Norm','None') into (1,1,0,-99)
    2) 'max_glu_serum' = replace('>200','>300','Norm','None') into (1,1,0,-99)

3. Outliers Detection and Treatment (3 Marks)

- Explore the data to identify potential outliers using **box plot, scatter plot and so on.**
- You are also allowed to demonstrate the outlier detection using DBSCAN, Interquartile Range (IQR) and Z-Score – choose any ONE method.
- Identify and visualize the potential outliers in each column.
- Decide on an outlier treatment strategy and apply it to the dataset.
- Handle the potential outliers and show the result after filtering/removing the outliers, preferably show it as ‘Before & After’.

Part 2- Feature Selection (5 Marks)

Please perform the following tasks using **Cleaned Dataset**:

1. **Perform Feature Selection as follows:**

- Select only the numerical columns to test for correlation- using a heatmap, plot the outcome. Explain the correlation between the features, which one is the best to select.

- Apply a forward selection by using linear regression, set k_features = 5. Record the selected features in form of bar graph.

- Apply an embedded method by using Lasso Regression. Record the selected features in form of bar graph.

Part 3- Data Visualisation (5 Marks)

Select any THREE numerical variables from **Cleaned Dataset**. Use Pandas to create a histogram, bar graphs, pie chart, heatmap to visualise the distribution of each of these variables. Briefly explain each graphs briefly.

C. Submission (Softcopy):

Each group should upload one submission to eBwise before the deadline. Your submission should be a **single normal zipped file**. The zipped file should include the following components:

- Completed **Group Assignment Declaration Form**. Failure to submit the completed form will result in a deduction of marks.
- Use **Lab Assignment_Part1_2_3 Template.ipynb (Jupyter Notebook file)** as a documentation template, it helps you to clearly indicate which questions you are answering in the Python files. Copy paste the codes accordingly and ensure the template file is free of error. Ensure the code can run and generate plots. Document your coding and explain the charts accordingly.

It is your responsibility to ensure that your submission file is correct and can be opened, the codes can be run without any errors. Please note that **resubmission after the assignment deadline is not permitted** for erroneous file submission.

D. Submission Date:

Please submit your assignment before **15 June 2026 (10 am).** Penalty will be imposed for late submission. 5% will be deducted for each overdue day.

E. BEWARE of Plagiarism

If you take an illustration or more than a few words of text from a book or other source, you must quote it and provide the source. Using the words or pictures of others without explicitly acknowledging them is plagiarism, a serious violation of scientific ethics. **5 marks will be deducted if plagiarism is found.**